24/9/26

Exp no: 07

SLIDING WINDOW PROTOCOL

Aim: Write a program to implement flow control at data link layer using SLIDING WINDOW PROTOCOL.

Program:

```
import time
message = input ("Enter the Text message: ")
window_size = int (input ("Enter window
                          size: "))
error-frame=int (input("Enter frame:"))
if window-size <=0:
     print ("Invalid window size")
     exit()
print(" SLIDING WINDOW PROTOCOL")
print ("\nMessage:", message)
print ("Window size:", window-size)
start =0
expected-frame =1
error-induced= False
while start< len(message):
    end=min (start+window-size, len(message))
    print (" SENDER BUFFER")
    Sender-buffer=[]
    for i in range(start, end):
        frame-no = i+1
        data= message[i]
    sender_buffer.append([frame_no, data])
    print("FRAME_NO =", frame_no, "DATA =", data)

print("\nSending frames...")
time.sleep(1)
print("    RECEIVER")
error_found = False
for frame in sender_buffer:
    frame_no = frame[0]
    data = frame[1]

    if frame_no == error_frame and not
                error_induced:
        received_frame = frame_no +1
        error_induced = True
    else:
        received_frame = frame_no

    print(
        "Received Frame:",
        received_frame,
        " DATA:",
        data
    )
    if received_frame == expected_frame:
        print("Frame", received_frame,
                        "is correct")

        expected_frame +=1

    else:
        print("\n ERROR DETECTED")
        print("Expected Frame:", expected_frame)
        print("Received Frame:", received_frame)
        print("  RECEIVER  BUFFER ")
        print("NACK = ", expected-frame)
        print("\nNACK received by Sender")
        print( "Resending frames from frame",
                  expected-frame)
        error-found = True
        break
if error-found:
    start = expected-frame -1
    time. spleep(1)
    continue
print("   RECEIVER   BUFFER")
ack = expected-frame
print ("ACK = ", ack)
print(" \n ACK  received by Sender")
print(" Expected  ACK:", end +1)

 if  ack == end +1:
      print("\n ACK is correct")
      print ("Sliding window moves
                            forward")

      start = end

 else:
      print("\n Incorrect ACK")
      print ("Waiting for correct ACK")
```

```
print(" TRANSMISSION COMPLETED")
print("\n Original Message: ", message)
print(" All frames successfully
                        transmitted")
```

OUTPUT :

```
Enter the text message: hello
Enter window size :
        SLIDING WINDOW PROTOCOL
Message : hello
Window Size : 3
            SENDER BUFFER
FRAME_NO = 1      DATA = h
FRAME_NO = 2      DATA = e
FRAME_NO = 3      DATA = l
Sending frames...
               RECEIVER
Received Frame : 1  DATA : h
Frame 1 is correct
Received Frame : 2  DATA : e
Frame 2 is correct
Received Frame : 8  DATA : l
            ERROR DETECTED
Expected Frame : 3
Received Frame : 8
```

```
            RECEIVER   BUFFER

NACK = 3
NACK received by Sender
Resending frames from Frame 3

            SENDER   BUFFER

FRAME_NO = 3      DATA = 1
FRAME_NO = 4      DATA = 1
FRAME_NO = 5      DATA = 0

Sending frames...

            RECEIVER

Received Frame: 3   DATA : 1
Frame 3 is correct
Received Frame: 4   DATA: 1
Frame 4 is correct
Received Frame: 5   DATA: 0
Frame is correct

            RECEIVER  BUFFER

ACK = 6

ACK received by Sender
Expected ACK : 6
ACK is correct
Sliding window moves forward
```

TRANSMISSION COMPLETED

Original Message: hello
All frames successfully transmitted.

Result: Thus flow control at data link layer using SLIDING WINDOW PROTOCOL.

```
TRANSMISSION COMPLETED

Original Message: hello
All frames successfully transmitted.
```

Result: Thus flow control at data link layer using SLIDING WINDOW PROTOCOL.

```
TRANSMISSION COMPLETED

Original Message: hello
All frames successfully transmitted.
```

Result: Thus flow control at data link layer using SLIDING WINDOW PROTOCOL.